PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

31. A stone of mass 900 g is tied to a string and moved in a vertical circle of radius 1 m making 10 rpm. The tension in the string, when the stone is at the lowest point is

(if $\pi^2 = 9.8$ and $g = 9.8$ m/s^2)

(1) 9.8 N (2) 8.82 N

(3) 97 N (4) 17.8 N

Answer (1)

Sol. $\omega = 10 \text{ rpm} = 10 \times \frac{\pi}{30} = \frac{\pi}{3}$ rad/s

$T - mg = m\omega^2 r$ $\quad m = 0.9$ kg, $r = 1$ m

$T - 0.9 \times 9.8 = 0.9 \times \frac{\pi^2}{9} \times 1$

$T = 0.9 \times 9.8 \left\{1 + \frac{1}{9}\right\}$

$= 0.9 \times 9.8 \times \frac{10}{9}$

$= 9.8$ N

32. The temperature of a gas having 2.0×10^{25} molecules per cubic meter at 1.38 atm

(Given, $k = 1.38 \times 10^{-23}$ JK^{-1}) is

(1) 300 K (2) 200 K

(3) 500 K (4) 100 K

Answer (3)

Sol. $PV = NkT$

$\Rightarrow P = \frac{N}{V}kT = nkT$

$\Rightarrow 1.38 \times 101325 = 2 \times 10^{25} \times 1.38 \times 10^{-23} T$

$\Rightarrow T = \frac{101325}{200} \approx 500$ K

33. If the distance between object and its two times magnified virtual image produced by a curved mirror is 15 cm, the focal length of the mirror must be

(1) –12 cm

(2) $\frac{10}{3}$ cm

(3) –10 cm

(4) 15 cm

Answer (3)

Sol. Let $|u| = x$

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ $\quad m = -\frac{v}{u}$

$\frac{1}{2x} - \frac{1}{x} = \frac{1}{f}$ $\quad 2 = \frac{-v}{-x}$

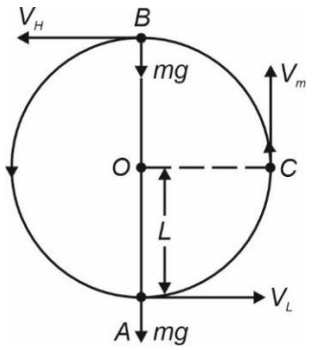
$-\frac{1}{2x} = \frac{1}{f}$ $\quad v = 2x$

$f = -2x$ $\quad$ Also $3x = 15 \Rightarrow x = 5$ cm

$f = -10$ cm



34. A bob of mass 'm' is suspended by a light string of length 'L'. It is imparted a minimum horizontal velocity at the lowest point A such that it just completes half circle reaching the top most positive B. The ratio of kinetic energies $\frac{(K.E)_A}{(K.E)_B}$ is



(1) 3 : 2 (2) 2 : 5
(3) 5 : 1 (4) 1 : 5

Answer (3)

Sol. $V_L = \sqrt{5gL}$ A lowest

$V_H = \sqrt{gL}$ B topmost

$$\frac{K_A}{K_B} = \frac{\frac{1}{2}m\,5gL}{\frac{1}{2}mgL} = \frac{5}{1}$$

35. A particle is moving in a straight line. The variation of position 'x' as a function of time 't' is given as $x = (t^3 - 6t^2 + 20t + 15)$ m. The velocity of the body when its acceleration becomes zero is

(1) 6 m/s (2) 10 m/s
(3) 8 m/s (4) 4 m/s

Answer (3)

Sol. $x = t^3 - 6t^2 + 20t + 15$

$\Rightarrow v = 3t^2 - 12t + 20$

$\Rightarrow a = 6t - 12$

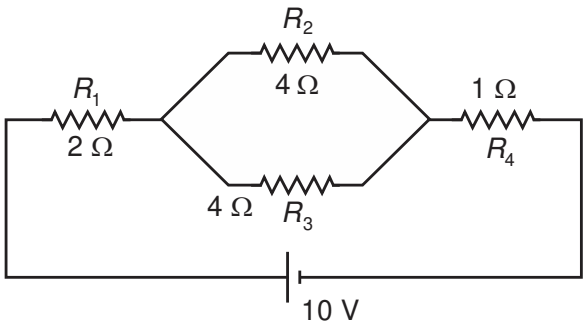
for $a = 0 \Rightarrow t = 2$

and $v_2 = 3 \times 2^2 - 12 \times 2 + 20$

$= 12 - 24 + 20$

$= 8$ m/s

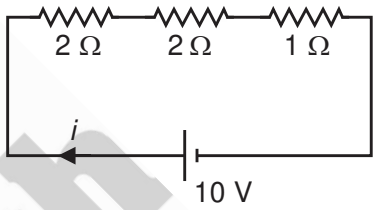
36. In the given circuit, the current in resistance R_3 is :



(1) 1.5 A (2) 1 A
(3) 2.5 A (4) 2 A

Answer (2)

Sol.



$$i = \frac{10}{5} = 2 \text{ A}$$

$$\Rightarrow i_3 = \frac{i}{2} = 1 \text{ A}$$

37. Two sources of light emit with a power of 200 W. The ratio of number of photons of visible light emitted by each source having wavelengths 300 nm and 500 nm respectively, will be :

(1) 1 : 5 (2) 3 : 5
(3) 5 : 3 (4) 1 : 3

Answer (2)

Sol. $P = \frac{\Delta n}{\Delta t}E_{Ph}$ and $E_{Ph} \propto \frac{1}{\lambda}$

$$\Rightarrow \frac{\Delta n_1}{\Delta t} \times \frac{1}{300} = \frac{\Delta n_2}{\Delta t} \times \frac{1}{500}$$

$$\Rightarrow \frac{\Delta n_1}{\Delta n_2} = \frac{3}{5}$$



38. A plane electromagnetic wave of frequency 35 MHz travels in free space along the *X*-direction. At a particular point (in space and time) $\vec{E} = 9.6\hat{j}$ V/m. The value of magnetic field at this point is :

(1) $9.6\hat{j}\,T$ (2) $9.6\times10^{-8}\hat{k}\,T$

(3) $3.2\times10^{-8}\hat{i}\,T$ (4) $3.2\times10^{-8}\hat{k}\,T$

Answer (4)

Sol. $E = cB$

$$\Rightarrow B = \frac{9.6}{3\times10^{8}} = 3.2\times10^{-8}\,\text{T}$$

39. *N* moles of a polyatomic gas ($f = 6$) must be mixed with two moles of a monoatomic gas so that the mixture behaves as a diatomic gas. The value of *N* is :

(1) 6 (2) 4

(3) 3 (4) 2

Answer (2)

Sol. $< f > = \frac{N_1 f_1 + N_2 f_2}{N_1 + N_2}$

$$\Rightarrow 5 = \frac{N6 + 2\times3}{N+2}$$

$\Rightarrow 5N + 10 = 6N + 6$

$\Rightarrow N = 4$

40. The bob of a pendulum was released from a horizontal position. The length of the pendulum is 10 m. If it dissipates 10% of its initial energy against air resistance, the speed with which the bob arrives at the lowest point is

[Use, g : 10 ms^{-2}]

(1) $6\sqrt{5}$ ms^{-1} (2) $5\sqrt{5}$ ms^{-1}

(3) $2\sqrt{5}$ ms^{-1} (4) $5\sqrt{6}$ ms^{-1}

Answer (1)

Sol. $\frac{9}{10}mg\ell = \frac{1}{2}mv^2$

$$\Rightarrow v = \sqrt{1.8g\ell}$$

$$= \sqrt{180}$$

$$= 6\sqrt{5}\,\text{m/s}$$

41. An electric field is given by $\left(6\hat{i}+5\hat{j}+3\hat{k}\right)$ N/C. The electric flux through a surface area $30\hat{i}$ m^2 lying in YZ-plane (in SI unit) is :

(1) 180

(2) 60

(3) 150

(4) 90

Answer (1)

Sol. $\phi_E = \vec{E}.\vec{A}$

$\Rightarrow \phi_E = 180$ SI unit

42. A planet takes 200 days to complete one revolution around the Sun. If the distance of the planet from Sun is reduced to one fourth of the original distance, how many days will it take to complete one revolution

(1) 20

(2) 25

(3) 50

(4) 100

Answer (2)

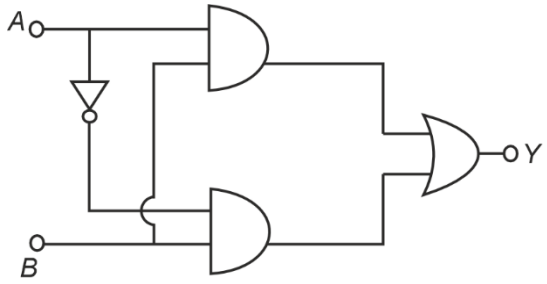
Sol. $T^2 \propto R^3$

$$\Rightarrow \frac{T_1}{T_2} = \frac{R^{\frac{3}{2}}}{\left(\frac{R}{4}\right)^{\frac{3}{2}}} = 8$$

$$\Rightarrow T_2 = \frac{200}{8} = 25$$



43. The truth table for this given circuit is :



(1)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(2)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(3)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(4)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

Answer (4)

Sol. $Y = A \cdot B + \overline{A} \cdot B$

$= \left(A + \overline{A}\right) B = B$

44. In an a.c. circuit, voltage and current are given by:

$V = 100 \sin (100\ t)$ V and

$I = 100 \sin \left(100t + \frac{\pi}{3}\right)$ mA respectively.

The average power dissipated in one cycle is :

(1) 25 W (2) 2.5 W

(3) 10 W (4) 5 W

Answer (2)

Sol. $P = V_{rms}\ I_{rms} \cos d$

$= \frac{100}{\sqrt{2}} \times \frac{100}{\sqrt{2}} \cos \frac{\pi}{3} \times 10^{-3}$

$= 2.5$ W

45. A physical quantity Q is found to depend on quantities a, b, c by the relation $Q = \frac{a^4 b^3}{c^2}$. The percentage error in a, b and c are 3%, 4% and 5% respectively.

Then, the percentage error in Q is :

(1) 14% (2) 34%

(3) 66% (4) 43%

Answer (2)

Sol. $\frac{\Delta Q}{Q} = \frac{4\Delta a}{a} + \frac{3\Delta b}{b} + \frac{2\Delta c}{c}$

% error = 4 × 3% + 3 × 4% + 2 × 5%

= (12 + 12 + 10)%

= 34%

46. A small liquid drop of radius R is divided into 27 identical liquid drops. If the surface tension is T, then the work done in the process will be:

(1) $\frac{1}{8}\pi R^2 T$ (2) $8\pi R^2 T$

(3) $4\pi R^2 T$ (4) $3\pi R^2 T$

Answer (2)

Sol. $27 \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$

$\Rightarrow R = 3r$

$\therefore W = \Delta V = 27 \times T \times 4\pi \left(\frac{R}{3}\right)^2 - T \times 4\pi R^2$

$T4\pi R^2 \times 2$

$W = 8\pi R^2 T$



47. A wire of length L and radius r is clamped at one end. If its other end is pulled by a force F, its length increases by l. If the radius of the wire and the applied force both are reduced to half of their original values keeping original length constant, the increase in length will become:

(1) 3 times (2) $\frac{3}{2}$ times

(3) 2 times (4) 4 times

Answer (3)

Sol. $\Delta l = \frac{F l}{YA}$

$\frac{\Delta l_1}{\Delta l_2} = \frac{F_1}{A_1}\frac{A_2}{F_2}$

$= \frac{2}{4} = \frac{1}{2}$

48. Two particles X and Y having equal charges are being accelerated through the same potential difference. Thereafter they enter normally in a region of uniform magnetic field and describes circular paths of radii R_1 and R_2 respectively. The mass ratio of X and Y is :

(1) $\left(\frac{R_1}{R_2}\right)^2$ (2) $\left(\frac{R_2}{R_1}\right)$

(3) $\left(\frac{R_1}{R_2}\right)$ (4) $\left(\frac{R_2}{R_1}\right)^2$

Answer (1)

Sol. $qV = K$

and, $r = \frac{\sqrt{2mk}}{qB}$

$\Rightarrow \frac{r_1}{r_2} = \frac{\sqrt{m_1}}{\sqrt{m_2}} = \frac{R_1}{R_2}$

$\Rightarrow \left(\frac{R_1}{R_2}\right)^2 = \frac{m_1}{m_2}$

49. In Young's double slit experiment, light from two identical sources are superimposing on a screen. The path difference between the two lights reaching at a point on the screen is $7\lambda/4$. The ratio of intensity of fringe at this point with respect to the maximum intensity of the fringe is :

(1) $\frac{1}{3}$ (2) $\frac{1}{4}$

(3) $\frac{3}{4}$ (4) $\frac{1}{2}$

Answer (4)

Sol. $I = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$

$\Delta\phi = \frac{2\pi}{\lambda} \times \frac{7\lambda}{4} = \frac{7}{2}\pi \equiv \frac{3}{2}\pi$

$\Rightarrow \frac{I}{I_0} = \cos^2\frac{3}{4}\pi = \frac{1}{2}$

50. Given below are two statements:

Statement I : Most of the mass of the atom and all its positive charge are concentrated in a tiny nucleus and the electrons revolve around it, is Rutherford's model.

Statement II : An atom is a spherical cloud of positive charges with electrons embedded in it, is a special case of Rutherford's model.

In the light of the above statements, choose the ***most appropriate*** from the options given below

(1) Both statement **I** and statement **II** are false

(2) Both statement I and statement **II** are true

(3) Statement **I** is true but statement **II** is false

(4) Statement **I** is false but statement **II** is true

Answer (3)

Sol. Rutherford model gave the planetary model of atom.

Statement I is correct and statement II is incorrect.



SECTION - B

Numerical Value Type Questions: This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

51. A charge of 4.0 μC is moving with a velocity of 4.0×10^6 ms^{-1} along the positive *y*-axis under a magnetic field $\vec{B}$ of strength $\left(2\hat{k}\right)$T. The force acting on the charge is $x\hat{i}$N. The value of *x* is _____.

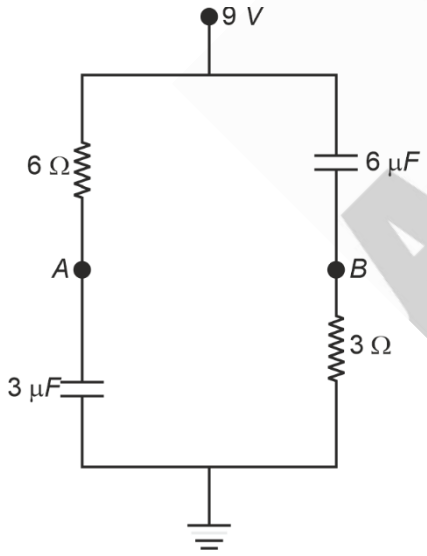
Answer (32)

Sol. $F = \left|q\left(\vec{v}\times\vec{B}\right)\right|$

$= 4\times10^{-6}\times4\times10^{6}\times2$

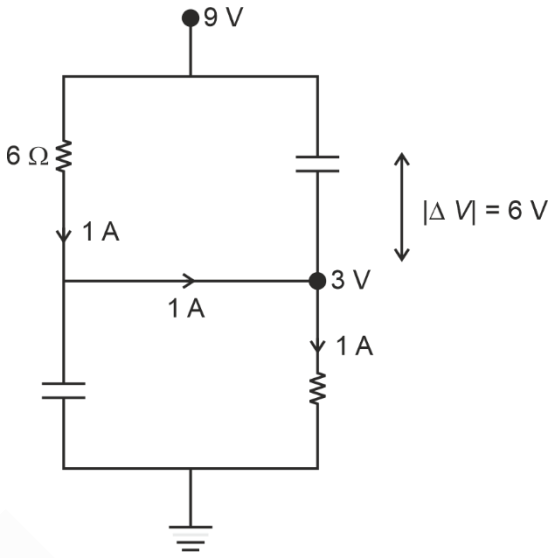
$= 32$

52. In the given figure, the charge stored in 6 μF capacitor, when point *A* and *B* are joined by a connecting wire is _____ μC.



Answer (36)

Sol.

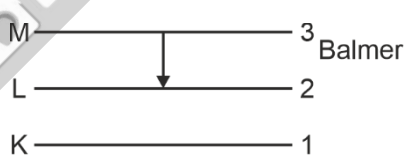


$q = 6 \times 6$ μC $= 36$ μC

53. Hydrogen atom is bombarded with electrons accelerated through a potential difference of *V*, which causes excitation of hydrogen atoms. If the experiment is being performed at *T* = 0 K, the minimum potential difference needed to observe any Balmer series lines in the emission spectra will be $\frac{\alpha}{10}$ V, where α = _____.

Answer (121)

Sol.



$|\Delta E| = 13.6\left\{1-\frac{1}{9}\right\}$

$= 12.08 = \frac{\alpha}{10}$

α = 121

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54. A horizontal straight wire 5 m long extending from east to west falling freely at right angle to horizontal component of earths magnetic field 0.60×10^{-4} Wbm^{-2}. The instantaneous value of emf induced in the wire when its velocity is 10 ms^{-1} is _____ $\times 10^{-3}$ V.

Answer (3)

Sol. $\varepsilon = Blv$

$= 0.6 \times 10^{-4} \times 5 \times 10$

$\varepsilon = 3 \times 10^{-3}$ V

55. Two metallic wires P and Q have same volume and are made up of same material. If their area of cross sections are in the ratio 4 : 1 and force F_1 is applied to P, an extension of Δl is produced. The force which is required to produce same extension in Q is F_2.

The value of $\frac{F_1}{F_2}$ is _____.

Answer (16)

Sol. $\Delta l = \frac{Fl}{YA} = \frac{FV}{YA^2}$

$\frac{\Delta l_1}{\Delta l_2} = \frac{F_1}{A_1^2}\frac{A_2^2}{F_2}$

$\frac{F_1}{16F_2} = \frac{\Delta l}{\Delta l}$

$\frac{F_1}{F_2} = 16$

56. A simple harmonic oscillator has an amplitude A and time period 6π second. Assuming the oscillation starts from its mean position, the time required by it to travel from $x = A$ to $x = \frac{\sqrt{3}}{2}A$ will be $\frac{\pi}{x}$ s, where x = _____ .

Answer (2)

Sol. $x = A \quad x = \frac{\sqrt{3}}{2}A$

$\Rightarrow \theta = 30°$

$t = \frac{T}{12} = \frac{6\pi}{12} = \frac{\pi}{2} = \frac{\pi}{x} \Rightarrow x = 2.$

57. A particle is moving in a circle of radius 50 cm in such a way that at any instant the normal and tangential components of it's acceleration are equal. If its speed at $t = 0$ is 4 m/s, the time taken to complete the first revolution will be $\frac{1}{\alpha}[1 - e^{-2\pi}]$ s, where α = _____ .

Answer (8)

Sol. $\frac{v^2}{r} = \left|\frac{dv}{dt}\right| \Rightarrow \frac{dt}{r} = \left|\frac{dv}{v^2}\right| \Rightarrow \frac{t}{r} = \left|\frac{1}{v_0} - \frac{1}{v}\right|$

$\frac{v^2}{r} = \left|\frac{vdv}{dx}\right| \Rightarrow \left|\frac{dv}{v}\right| = \frac{dx}{r} \Rightarrow \left|\ln\frac{v}{v_0}\right| = \frac{s}{r} = \frac{2\pi r}{r}$

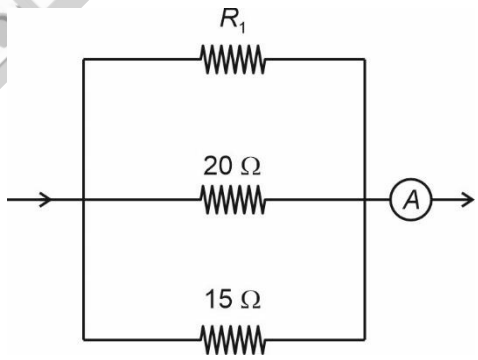
$\Rightarrow \frac{v}{v_0} = e^{\pm 2\pi}$

$v = v_0 e^{\pm 2\pi}$

$\frac{t}{r} = \left|\frac{1}{v_0} - \frac{1}{v_0 e^{\pm 2\pi}}\right|$

$\Rightarrow t = \frac{1}{8}\left[1 - e^{-2\pi}\right] = \frac{1}{\alpha}[1 - e^{-2\pi}] \Rightarrow \alpha = 8$

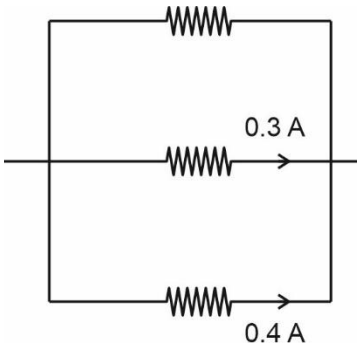
58. In the given circuit, the current flowing through the resistance 20 Ω is 0.3 A, while the ammeter reads 0.9 A. The value of R_1 is _____ Ω.



Answer (30)

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Sol. $V_{R_1} = V_{20} = V_{15} = 0.3 \times 20 = 6$ V



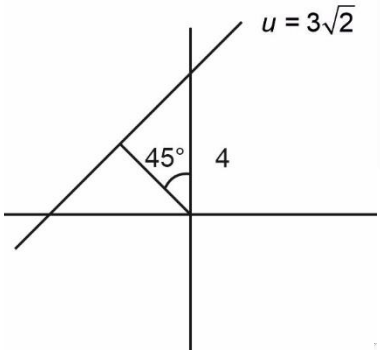
$\Rightarrow \quad i_1 = 0.2$ A

$$R_1 = \frac{6}{0.2} = 30\ \Omega$$

59. A body of mass 5 kg moving with a uniform speed $3\sqrt{2}$ ms^{-1} in X - Y plane along the line $y = x + 4$. The angular momentum of the particle about the origin will be _____ kg m^{2}s^{-1}.

Answer (60)

Sol.



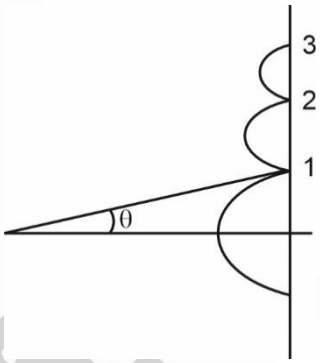
$L = mvr_1$

$$= 5 \times 3\sqrt{2} \times \frac{4}{\sqrt{2}} = 60 \text{ SI unit}$$

60. In a single slit diffraction pattern, a light of wavelength 6000 Å is used. The distance between the first and third minima in the diffraction pattern is found to be 3 mm when the screen in placed 50 cm away from slits. The width of the slit is _____ $\times 10^{-4}$ m.

Answer (2)

Sol. $d\sin\theta = n\lambda$ for minima



$$\theta_1 = \frac{\lambda}{d}$$

$$\theta_3 = \frac{3\lambda}{d}$$

$$\Delta\theta = \frac{2\lambda}{d}$$

$$D\Delta\theta = \frac{2\lambda}{d} D = 3\text{mm} = \frac{2 \times 6000 \times 10^{-10}}{d} \times \frac{1}{2}$$

$$\Rightarrow \quad d = \frac{6 \times 10^{-7}}{3 \times 10^{-3}} = 2 \times 10^{-4} \text{ m}$$



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. The element having the highest first ionization enthalpy is

(1) C (2) N

(3) Al (4) Si

Answer (2)

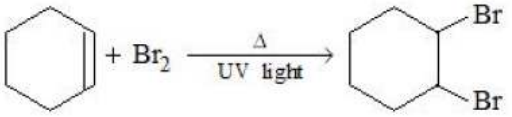
Sol. In general ionisation enthalpy increases on moving left to right in the period and decreases on moving top to bottom in group. Hence 1st ionisation enthalpy is highest for N.

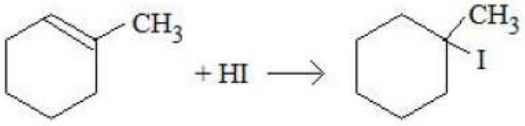
62. Which of the following reaction is correct?

(1) $C_2H_5CONH_2 + Br_2 + NaOH \longrightarrow$

$C_2H_5CH_2NH_2 + Na_2CO_3 + NaBr + H_2O$

(2) $CH_3CH_2CH_2NH_2 \xrightarrow[H_2O]{HNO_2,\ 0^oC} CH_3CH_2CH_2OH + N_2 + HCl$

(3) 

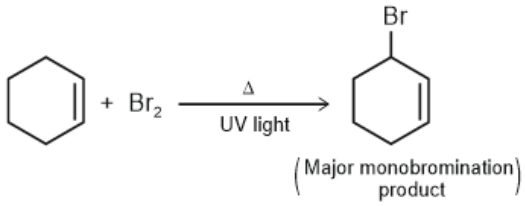
(4) 

Answer (4)

Sol. The correct organic products of given reactions are as follows

1. $C_2H_5CONH_2 + Br_2 + NaOH \rightarrow C_2H_5NH_2$

2. $CH_3CH_2CH_2NH_2 \xrightarrow[H_2O]{HNO_2,\ 0^oC} CH_3CH_2CH_2OH$

3. 

4. 

Hence option (4) is correct.

63. A reagent which gives brilliant red precipitate with Nickel ions in basic medium is

(1) Dimethyl glyoxime

(2) Meta-dinitrobenzene

(3) Neutral $FeCl_3$

(4) Sodium nitroprusside

Answer (1)

Sol. Ni^{2+} + dimethyl glyoxime $\xrightarrow{\text{Basic medium}}$ $[Ni(dmg)_2]$ (Cherry red or brilliant red)

64. The correct IUPAC name of K_2MnO_4 is

(1) Potassium tetraoxopermanganate (VI)

(2) Potassium tetraoxidomanganate (VI)

(3) Dipotassium tetraoxidomanganate (VII)

(4) Potassium tetraoxidomanganese (VI)

Answer (2)

